



AI IN HEALTHCARE TRAINING PROGRAM

**10 WEEKS | WORLD CLASS INDUSTRY TRAINERS
GLOBAL IMPRESSION**

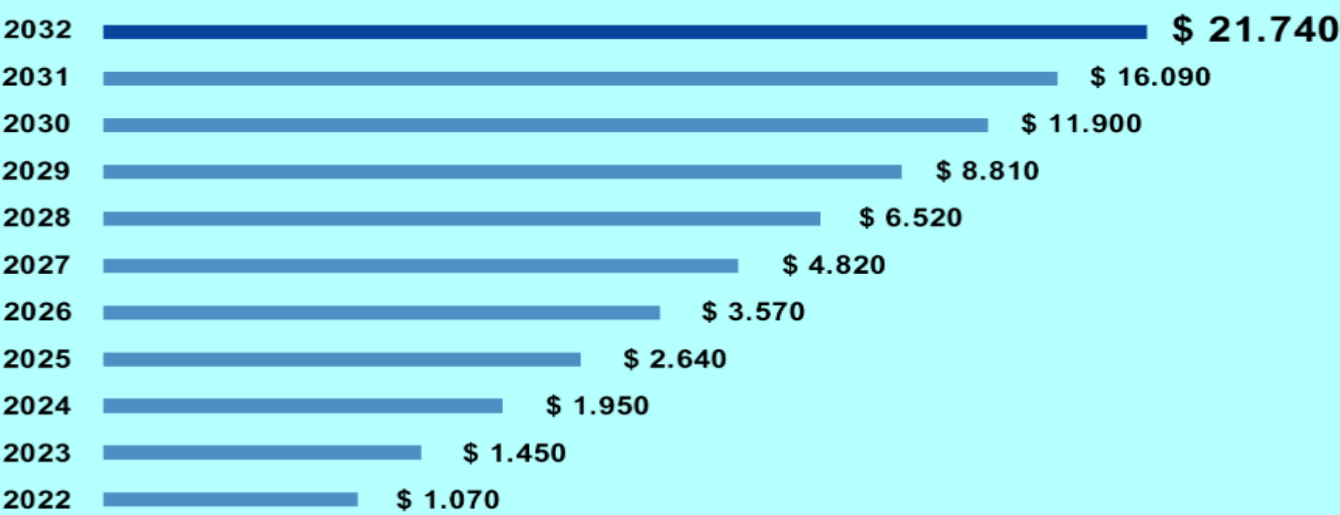


Generative AI

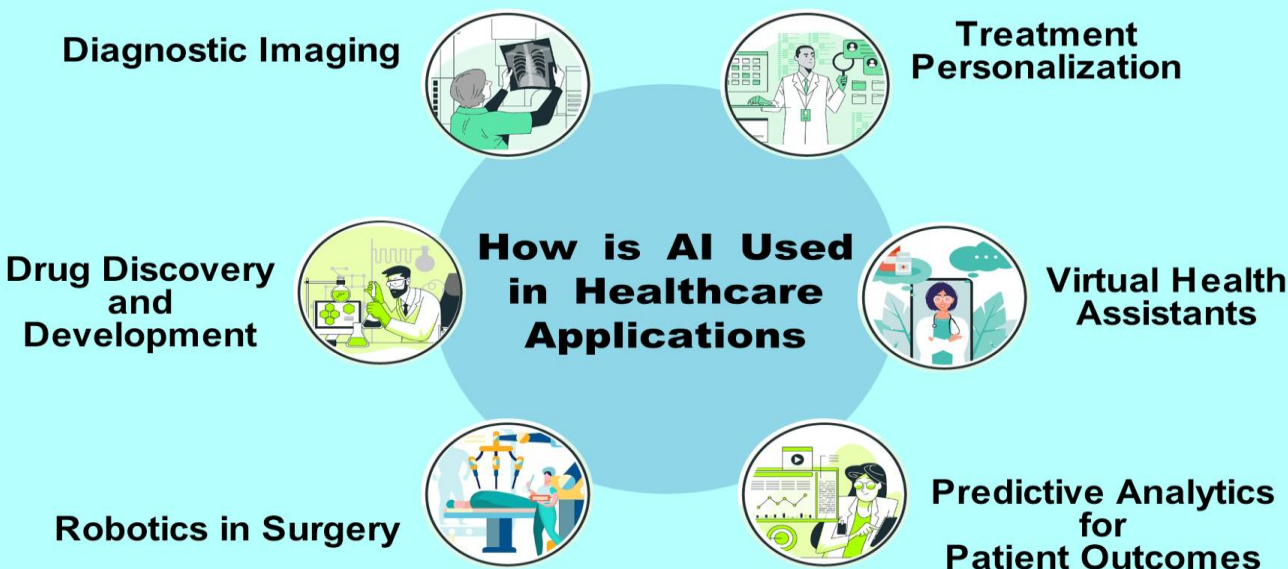
Revolutionizing Healthcare Operations

Artificial intelligence has remodeled the healthcare landscape in the past few years delivering breakthroughs in medical diagnostics, planning of treatment among other areas of patient care. With the amount and caliber of AI-powered solutions continuing to expand, there's a realignment of what's possible in medicine by making once futuristic ideas become actualized.

**Generative AI in Healthcare Market Revenue
2022 to 2032 (USD Million)**



The global artificial intelligence in the healthcare market was valued at USD 1,070 million in 2022 and is estimated to hit around USD 21,740 million by 2032, growing at a healthy CAGR of 35.1% from 2023 to 2032.



ABOUT THE PROGRAM

<u>Week #</u>	<u>Objective</u>	<u>Topics</u>
<u>Week-1</u> The Pharma Industry Meets Data Science	Understand the pharma ecosystem and why data is its new currency	1 - <i>How a Drug Gets to Market:</i> From Discovery to Commercialization 2 - <i>Key Data Types:</i> Introduction to Claims Data, Electronic Health Records (EHRs), Clinical Trial Data, and Sales Data 3 - <i>Hands-On:</i> No-code exploration of a sample dataset in Excel/Power BI to find a simple insight. 4 - <i>Career Spotlight:</i> What do a Data Scientist, Bioinformatician, and Market Analyst actually do in pharma?
<u>Week-2</u> Python Crash Course	Build foundational coding skills in a supportive environment.	1 - Python Basics 2 - Advance Libraries 3 - Lab 1: Loading a CSV file, cleaning messy data, and creating a simple bar chart 4 - Focus: "Getting your hands dirty"
<u>Week-3</u> SQL	Learn to extract the patient level data	1 - SQL Fundamentals 2 - Lab 2: Using a web-based SQL platform (like MySQL in the browser) to query a sample patient database to find the most prescribed drug in a region. 3 - Project Kick-off: Form teams and choose a final project theme.
<u>Week-4</u> The Art of Visualization - Telling Stories with Data	Communicate findings effectively through PowerBI dashboards	1 - Principles of Effective Data Visualization 2 - Introduction to Power BI/Tableau: Connecting to data and building interactive charts. 3 - Lab 3: Building a simple dashboard to track disease prevalence over time and by region. 4 - Tool: Power BI Desktop (Free).
<u>Week-5</u> Forecasting - Predicting the Future	Understand the basic concepts of predicting trends	1 - What is Forecasting? Time Series patterns, seasonality. 2 - Simple Models: Moving Averages, Linear Regression, Arima, Monte Carlo Simulation 3 - Lab 4: Using Python to forecast future sales of a product based on its past performance 4 - Context: Why forecasting is critical for drug launches and inventory management

ABOUT THE PROGRAM (Cont...)

Introduction to AI & Machine Learning	build your first ML model	2 - Your First ML Model: Using `scikit-learn` to build a model that predicts patient adherence based on simple features 3 - Lab 5: A step-by-step, guided lab to build and evaluate a simple classification model 4 - Focus: Understanding the process, not the complex math & stats
<u>Week-7</u> : Patient-Level Data (APLD) - The Goldmine	Work with the most important type of pharma data	1 - What is APLD? Understanding de-identified patient journeys 2 - Ethical Use of Healthcare Data: Privacy, anonymity, and HIPAA/GDPR 3 - Lab 6: Analyzing a sample APLD dataset to map the patient treatment pathway for a disease like diabetes or hypertension
<u>Week-8</u> Market Access & Commercial Analytics Simplified	See how analytics drives business strategy	1 - How Do Drugs Get Paid For? Intro to Payers, Pricing, and Reimbursement 2 - Commercial Analytics: How companies measure the impact of their sales and marketing teams 3 - Lab 7: Calculating market share and sales growth from a sample dataset for a portfolio of products
<u>Week-9</u> Capstone Project Work Week - Bring It All Together	Apply all skills to a fun, end-to-end project	1 - Sample Projects: "Analyze the launch of a new vaccine," "Optimize a sales rep's territory," "Build a dashboard for a hospital's drug spend." 2 - Dedicated team work time with heavy mentor and TA support 3 - Focus on the process: Data Extraction (SQL) -> Analysis (Python) -> Storytelling (Power BI Dashboard + Presentation)
<u>Week-10</u> Final Presentations & Career Launchpad	Showcase work and learn how to get a job in the field	1 - Capstone Fair: Teams present their projects to instructors and peers 2 - Building Your Resume: How to list projects and skills to stand out to recruiters 3 - Panel Discussion 4 - Certificate of Completion Awarded

